TopBike New System Proposal

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Table of Contents

Part 1 - Project Scope and Feasibility **4**
Business and System Context Analysis **4**
Company Overview 4
Project Overview 4
System Proposal 4
System Feasibility 4
System Request **4**
Project Sponsor 5
Business Need 5
Business Requirements 5
Business Value 5
Special Issues or Constraints 5
Feasibility Analysis Report **6**
Technical Feasibility 6
Economic Feasibility 6
Operational Feasibility 6
Conclusion 6
Cost-Benefit Analysis **7**
Overview 7
ROI and Break-even Point 7
Conclusion 7
Part 2 - Systems Requirements **8**
Requirement Definition **8**
Functional Requirements 8
Non-Functional Requirements 8
Use Case Analysis **9**
Use Case 1 9
Use Case 2 11
Use Case 3 14
Data Flow Diagrams (DFDs) **17**
Context Diagram 17
Level 0 Diagram 17
Lower-level Diagrams (Level 1) 18
Part 3 - Data Models **20**
Entity-Relationship Diagram **20**
Diagram 20
Overview 20
Model Breakdown **21**
Entities and Attributes 21
Relationships 21
Primary and Foreign Key Validation 22

Normalization 22
CRUD Matrix Table **23**
Part 4 - User Interface Design **24**
User Interface **24**
User Profiles 24
Interface Structure Diagram 25
UI Prototypes 26
Physical Data Flow Diagrams **29**
Structure Chart **30**
Part 5 - Test Plan **31**
Physical ERD **31**
Diagram 31
Test Plans **32**
Point-of-Sale (POS) System 32
Online Ordering System 34
Invoice Generation System for Trade Customers 36

Part 1 - Project Scope and Feasibility

Business and System Context Analysis

Company Overview

TopBike is a bicycle shop owned and managed by Herbert, offering a range of products including bicycles, mopeds, accessories, and repair services. The business also fulfills telephone orders for both individual customers and trade clients. Due to increasing competition from large retailers like Malfords, Herbert has secured a £400,000 loan to expand his shop, doubling its floor space. The goal is to increase annual revenue from £800,000 to £1,600,000 within three years. A key component of this expansion is improving the outdated manual information system, which is essential for managing the anticipated increase in sales and inventory.

Project Overview

Herbert has commissioned a £2,000 feasibility study to analyze TopBike's current system and explore modernization options. The existing processes rely heavily on manual record-keeping, leading to inefficiencies in stock management, order processing, and financial tracking. Key issues include inaccurate stock records, delayed invoicing, discount fraud, and limited tracking of outstanding payments, which total approximately £40,000.

System Proposal

To address these challenges, one of TopBike's employees, Ged, has proposed a new IT system, which includes two PCs, barcode scanners, and a web-based ordering platform. The estimated cost is £10,000 for hardware and an additional £5,000 annually for Ged's salary increase. However, Herbert lacks technical expertise to evaluate this solution.

System Feasibility

Our team will conduct a feasibility study to determine whether Ged's system is a viable option or if an alternative, such as a commercial off-the-shelf (COTS) solution or a cloud-based inventory management system, would better suit TopBike's needs. This project aims to develop a logical model of the current system, identify inefficiencies, and propose an improved system that aligns with TopBike's expansion goals while ensuring operational efficiency and financial stability.

System Request

Project Sponsor

Herbert, Owner/Manager of TopBike

Business Need

TopBike is expanding to compete with large retailers. Herbert secured a £400,000 loan, but the manual system is inefficient. A computerized system is needed to improve operations and double turnover in three years.

Business Requirements

- Automate stock control to ensure accurate inventory tracking and prevent stock discrepancies.
- Implement a point-of-sale system to record sales in real-time and integrate with stock updates.
- Digitize customer and trade orders to improve order tracking and reduce processing delays.
- Introduce an online ordering system via a website to expand market reach and increase sales.
- Automate invoice tracking and payment reminders to reduce outstanding debts.
- Implement security controls to prevent unauthorized discounts and potential stock theft

Business Value

- Improves operational efficiency by reducing manual record-keeping.
- Minimizes stock discrepancies, reducing over/under-ordering.
- Speeds up customer transactions with real-time sales recording.
- Increases revenue by allowing online orders.
- Enhances customer satisfaction through better order fulfillment and tracking.
- Reduces outstanding debts by automating payment reminders.

Special Issues or Constraints

- The system must be operational within two months to support business growth.
- Budget of £10,000 for hardware and initial setup.
- Ongoing costs include £5,000 per year for system maintenance and £400 per year for website hosting.
- The feasibility of using Ged’s proposed system versus a professionally developed system should be evaluated.
- System must be user-friendly for non-technical staff.

Feasibility Analysis Report

Technical Feasibility

TopBike's proposed information system is technically feasible. The company has access to modern computing resources, cloud-based storage, and experienced IT personnel who can implement and maintain the system. The software required for inventory management, customer relations, and order processing is readily available, either as an off-the-shelf solution or through custom development. Integration with existing hardware and online platforms should be straightforward, minimizing technical risks.

Economic Feasibility

The financial viability of the system is strong. While the initial investment in software, hardware, and training will require capital, the long-term benefits outweigh the costs. Improved operational efficiency, reduced manual errors, and enhanced customer service will lead to higher revenue and lower overhead costs. Additionally, potential increases in online sales and improved customer retention further justify the investment. Conducting a cost-benefit analysis confirms that the return on investment (ROI) is favorable.

Operational Feasibility

The system aligns with TopBike's business objectives and operational needs. Employees will require some training, but the learning curve is manageable. The new system will enhance inventory tracking, streamline customer interactions, and optimize supply chain management, leading to smoother business operations. Moreover, user adoption is expected to be high due to the system's ease of use and clear benefits in daily tasks.

Conclusion

Based on the technical, economic, and operational feasibility analysis, the proposed information system for TopBike is viable and should proceed to the next phase of development.

Cost-Benefit Analysis

Overview

COMPANY NAME	TOPBIKE	DATE CONDUCTED	1/28/2025 – 2/7/2025
PROPOSED INITIATIVE	NOASTHMA REBRANDING	COMPLETED BY	1/28/2025 – BILL GERLACH

COSTS	YEAR 0	YEAR 1	YEAR 2	YEAR 3	YEAR 4	YEAR 4	TOTAL
DEVELOPMENT COSTS							
Computer System (2 PCs, Peripherals)	£10,000.00	-	-	-	-	-	£10,000.00
Website Development	Included in system	-	-	-	-	-	£0.00
Feasibility Study	£2,000.00	-	-	-	-	-	£2,000.00
TOTAL DEVELOPMENT COSTS	£12,000.00	£0.00	£0.00	£0.00	£0.00	£0.00	£12,000.00
OPERATIONAL COSTS							
Loan Repayment	-	£20,000.00	£20,000.00	£20,000.00	£20,000.00	£20,000.00	£100,000.00
Advertising Budget	-	£14,000.00	£14,000.00	£14,000.00	£14,000.00	£14,000.00	£70,000.00
IT Management (Salary Increase)	-	£5,000.00	£5,000.00	£5,000.00	£5,000.00	£5,000.00	£25,000.00
Annual ISP Cost	-	£400.00	£400.00	£400.00	£400.00	£400.00	£2,000.00
TOTAL OPERATIONAL COSTS	£0.00	£39,400.00	£39,400.00	£39,400.00	£39,400.00	£39,400.00	£197,000.00
TOTAL COSTS	£12,000.00	£39,400.00	£39,400.00	£39,400.00	£39,400.00	£39,400.00	£209,000.00

BENEFITS	YEAR 0	YEAR 1	YEAR 2	YEAR 3	YEAR 4	YEAR 4	TOTAL
REVENUES							
Online Sales	-	£100,000.00	£100,000.00	£100,000.00	£100,000.00	£100,000.00	£500,000.00
Reduced Losses (Stock Errors)	-	£33,783.00	£33,783.00	£33,783.00	£33,783.00	£33,783.00	£168,915.00
Improved Cash Flow (Faster Invoicing)	-	£32,000.00	£32,000.00	£32,000.00	£32,000.00	£32,000.00	£160,000.00
TOTAL REVENUES	£0.00	£165,783.00	£165,783.00	£165,783.00	£165,783.00	£165,783.00	£828,915.00
TOTAL BENEFITS	£0.00	£165,783.00	£165,783.00	£165,783.00	£165,783.00	£165,783.00	£828,915.00

ROI and Break-even Point

ANALYSIS	YEAR 0	YEAR 1	YEAR 2	YEAR 3	YEAR 4	TOTAL
BENEFITS						
REVENUE	£0.00	£165,783.00	£165,783.00	£165,783.00	£165,783.00	£828,915.00
TOTAL BENEFITS	**£0.00**	**£165,783.00**	**£165,783.00**	**£165,783.00**	**£165,783.00**	**£828,915.00**
COSTS						
DEVELOPMENT	£12,000.00	£0.00	£0.00	£0.00	£0.00	£12,000.00
OPERATIONAL	£0.00	£39,400.00	£39,400.00	£39,400.00	£39,400.00	£197,000.00
TOTAL COSTS	**£12,000.00**	**£39,400.00**	**£39,400.00**	**£39,400.00**	**£39,400.00**	**£209,000.00**
CUMULATIVE CASH FLOW	**-£12,000.00**	**£114,383.00**	**£240,766.00**	**£367,149.00**	**£493,532.00**	**£1,113,447.00**

NET PRESENT VALUE	£619,915.00
RETURN ON INVESTMENT	296.61%
BREAK-EVEN POINT (IN YEARS)	2.80

Conclusion

Because of the aggressive rebranding strategy and system improvements that TopBike plans to employ, it is projected that there will be a roughly 300% return on investment. Our team believes that Herbert should move forward with this plan, as the company stands to gain significant cash flow and long-term sustainability.

Part 2 - Systems Requirements

Requirement Definition

Functional Requirements

Stock Management Automation: The system must track inventory levels in real-time. Automatic updates should be made when stock is purchased, sold, or reordered. Low stock alerts must be generated when items reach a predefined threshold.

Point-of-Sale (POS) System: Sales must be recorded in real-time. The system should integrate with inventory updates. It must support multiple payment methods (credit card, cash, online payments).

Customer & Trade Order Processing: Orders must be digitized and tracked from initiation to fulfillment. The system must allow for customer order status tracking. The system should support bulk trade orders with specific discounts.

Online Ordering System: Customers must be able to place orders via the website. The system should provide live stock availability on the website. Order confirmations and updates should be sent via email.

Invoice Tracking & Payment Reminders: Automatic invoice generation for purchases. Payment reminders should be sent for outstanding balances. The system should track overdue payments and flag high-risk customers.

Security & Fraud Prevention: Role-based access control should be implemented to prevent unauthorized actions. Employee discount approvals must be logged and require manager authorization. Activity logs should be maintained for all system transactions.

Non-Functional Requirements

Usability & Accessibility: The system must have an intuitive user interface for non-technical staff. It should support multiple languages for diverse customers. The web interface must be mobile-responsive.

Performance & Scalability: The system should handle at least 500 transactions per hour without performance degradation. It must support at least a 2x increase in concurrent users within three years.

Reliability & Availability: The system should maintain at least 99.9% uptime. Daily backups should be performed to prevent data loss.

Security & Compliance: The system must use encryption for sensitive data. It must comply with GDPR and PCI DSS for payment security.

Maintainability & Support: The system should be modular for easy updates. An admin panel should allow system updates and troubleshooting.

Use Case Analysis

Use Case 1

Use Case Description		
Use Case Name: Processing a Customer Purchase	**ID:** 1	**Importance Level:** High
Primary Actor(s): Customer, Cashier **Short Description:** This describes how a customer selects products, the cashier processes the transaction, and the system updates inventory and records the sale.		
Trigger: Customer initiates a purchase at checkout.		
Type: External / Temporal		

Major Inputs	
Description	**Source**
Selected product(s)	Customer
Scanned barcode or manual entry	Cashier
Payment method selection	Customer

Major Outputs	
Description	**Destination**
Updated inventory	System
Sales transaction record	System
Printed receipt	Customer

Major Steps Performed	
1. Customer selects product(s) for purchase.	5. System processes payment.
2. Cashier initiates the checkout process.	6. System prints a receipt.
3. System displays the total price.	7. System updates inventory and records the transaction.
4. Customer selects and provides a payment method.	

Information for Steps	
Step Description	**Required Data**
Product selection	Product list
Checkout initiation	Selected products
Display total price	Product prices
Payment processing	Payment details
Inventory update	Product quantity
Transaction recording	Sales data

Use Case 2

Use Case Description		
Use Case Name: Placing an Online Order	**ID:** 2	**Importance Level:** High
Primary Actor(s): Customer, System **Short Description:** This describes how a customer selects items online, provides necessary details, and confirms the order, after which the system processes payment and updates stock.		
Trigger: Customer initiates an order on the website.		
Type: External / Temporal		

Major Inputs	
Description	**Source**
Selected items	Customer
Shipping and payment details	Customer
Order confirmation	Customer

Major Outputs	
Description	**Destination**
Stock updated	System
Order confirmation email sent	Customer
Payment processed	System

Major Steps Performed	
1. Customer logs into the website.	4. System verifies stock availability.
2. Customer selects items and adds them to the cart.	5. Payment is processed securely.
3. Customer proceeds to checkout and provides shipping/payment details.	6. Order is confirmed and queued for fulfillment.

Information for Steps	
Step Description	**Required Data**
Customer login	User credentials
Item selection and cart addition	Product list
Checkout process	Shipping/payment info
Stock verification	Inventory database
Payment processing	Secure transaction data
Order confirmation	Order details

Use Case 3

Use Case Description		
Use Case Name: Generating an Invoice for Trade Customers	**ID:** 3	**Importance Level:** High
Primary Actor(s): Trade Customer, System **Short Description:** This describes how an invoice is generated for trade customers after bulk purchases, with applicable discounts and recorded payment terms.		
Trigger: Trade customer places a bulk order.		
Type: External / Temporal		

Major Inputs:	
Description	**Source**
Selected products	Trade Customer
Bulk discount application (if applicable)	System
Payment terms agreement	Trade Customer

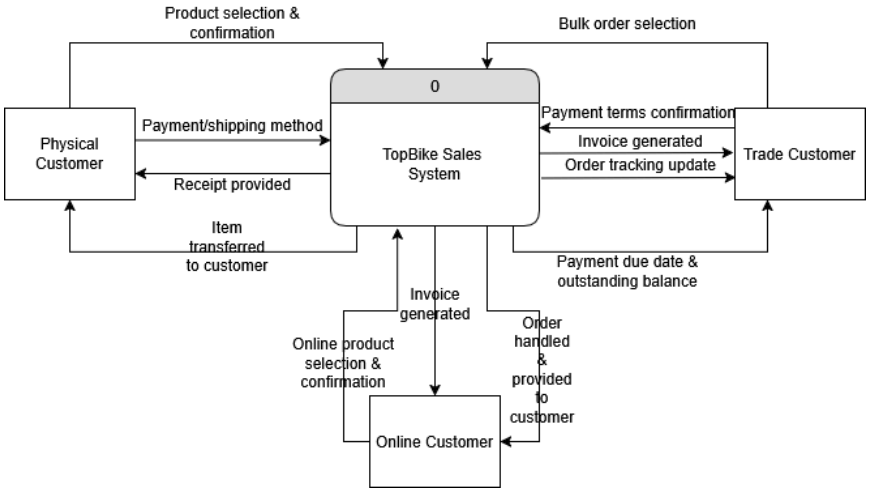
Major Outputs	
Description	**Destination**
Invoice generated	Trade Customer
Payment due date recorded	System
Outstanding balance updated	System

Major Steps Performed	
1. Trade customer places a bulk order.	4. Invoice is sent to the trade customer.
2. System calculates total price, applying bulk discounts if applicable.	5. Payment terms are recorded in the system.
3. System generates an invoice.	6. System updates outstanding balances.

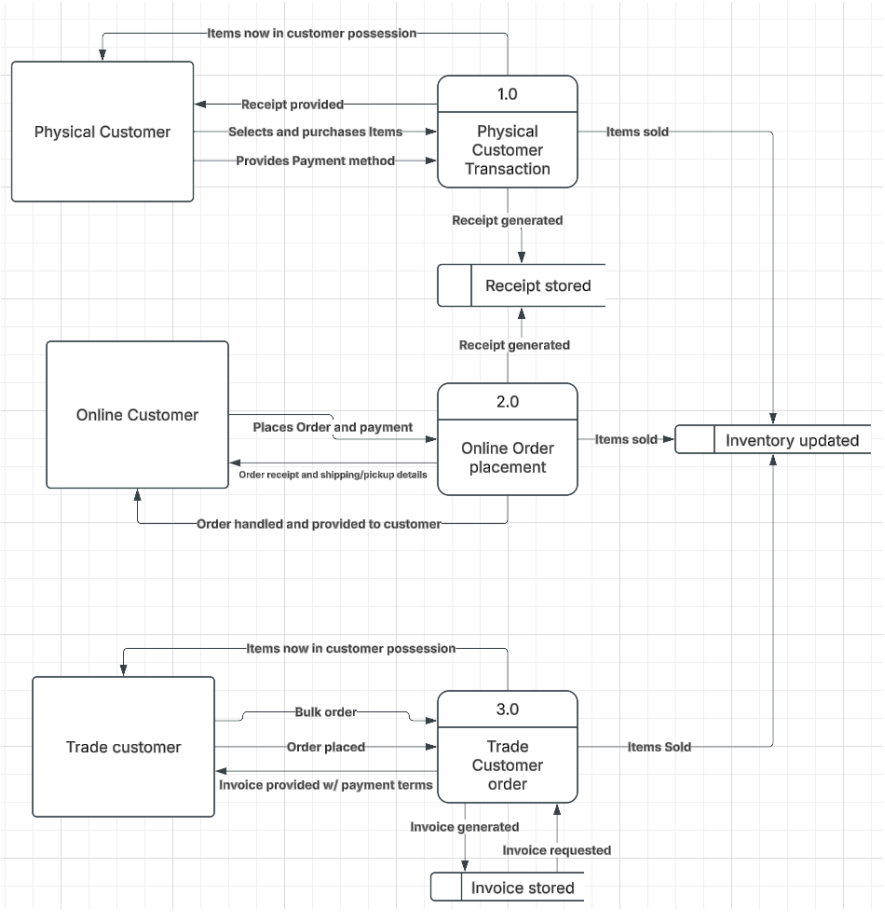
Information for Steps	
Step Description	**Required Data**
Bulk order placement	Selected products
Price calculation	Product prices, discount policies
Invoice generation	Order details, customer information
Invoice delivery	Customer contact details
Payment terms recording	Agreed payment terms
Outstanding balance update	System financial records

Data Flow Diagrams (DFDs)

Context Diagram

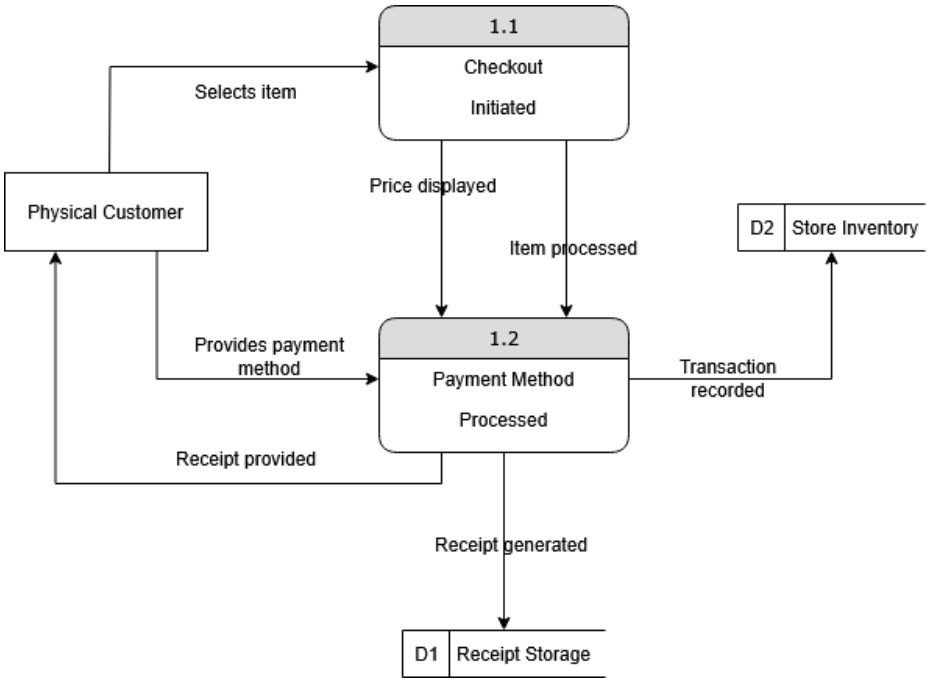


Level 0 Diagram

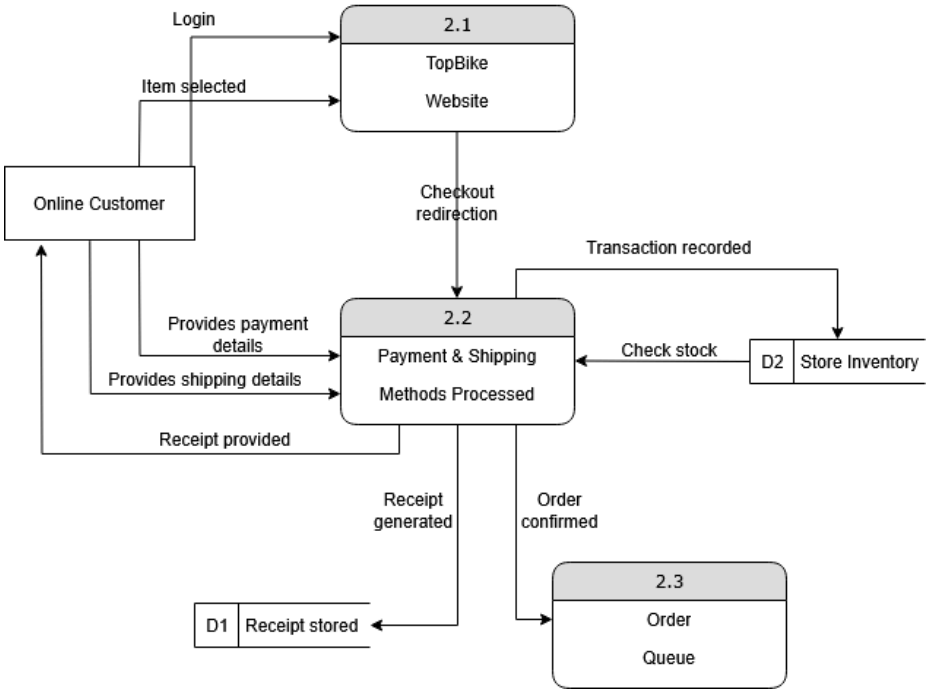


Lower-level Diagrams (Level 1)

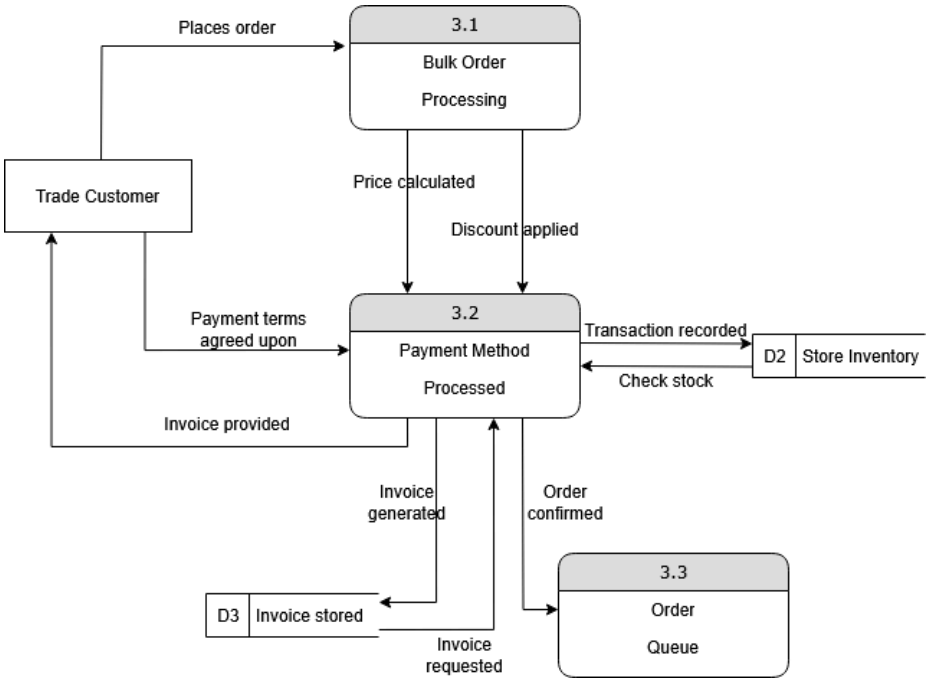
Process 1



Process 2



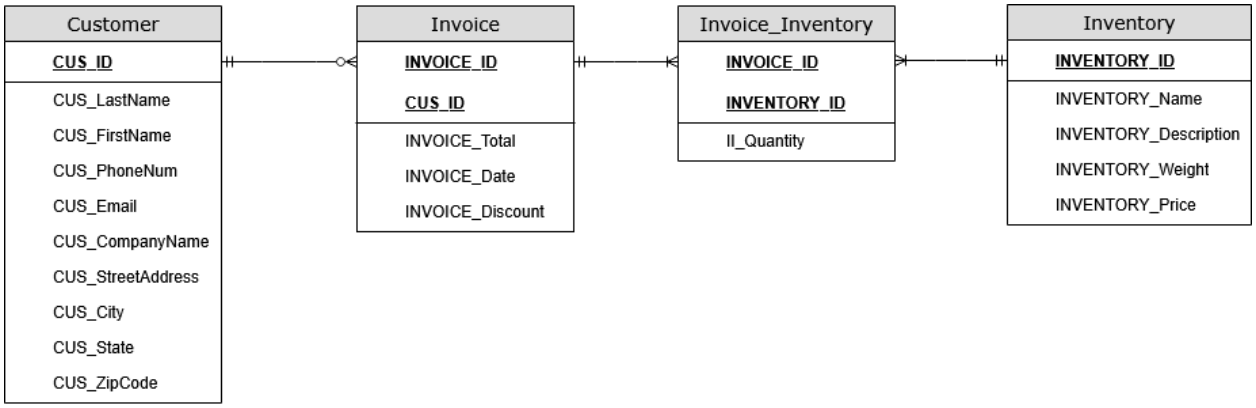
Process 3


Places order
3.1
Bulk Order
Processing
Trade Customer
Price calculated
Discount applied
Payment terms
agreed upon
3.2
Payment Method
Processed
Transaction recorded
D2
Store Inventory
Check stock
Invoice provided
Invoice
generated
Order
confirmed
3.3
Order
Queue
D3
Invoice stored
Invoice
requested

Part 3 - Data Models

Entity-Relationship Diagram

Diagram



Overview

Our level 0 data-flow diagram previously consisted of the external entities trade customer, physical customer and online customer. Here, we have simplified these into a single entity called “Customer”, because those entities in the DFD shared similar functions. Similarly, the “Invoice” entity condenses the data stores “Receipt stored” and “Invoice stored”.

In the second part of this project, we defined our system requirements which included bulk order processing and receipt distribution. To implement bulk order requests and processing, we have included an order quantity attribute in the “Invoice_Inventory” table, which will identify bulk orders at a certain threshold. The “INVOICE_Discount” attribute in the “Invoice” entity then applies a percentage-based discount if that threshold is passed.

Additionally, the Invoice entity in this ERD functions as both a receipt and invoice system, where it will tie order information such as “INVOICE_ID” and “INVOICE_Date” to the “Customer” entity, allowing us to track orders easily.

A final point of interest is the “Invoice_Inventory” table, which is a junction table created to bypass the many-to-many relationship between the “Invoice” table and the “Inventory” table.

Model Breakdown

Entities and Attributes

Customer

- **PK:** CUS_ID (Primary Key)
- **Other attributes:** CUS_LastName, CUS_FirstName, CUS_PhoneNum, CUS_Email, CUS_CompanyName, CUS_StreetAddress, CUS_City, CUS_State, CUS_ZipCode

Invoice

- **PK:** INVOICE_ID (Primary Key)
- **FK:** CUS_ID (Foreign Key from Customer)
- **Other attributes:** INVOICE_Total, INVOICE_Date, INVOICE_Discount

Invoice_Inventory (Associative/Bridge Entity)

- **FK1:** INVOICE_ID (Foreign Key from Invoice)
- **FK2:** INVENTORY_ID (Foreign Key from Inventory)
- **Other attributes:** II_Quantity (Quantity of items in the order)

Inventory

- **PK:** INVENTORY_ID (Primary Key)
- **Other attributes:** INVENTORY_Name, INVENTORY_Description, INVENTORY_Weight, INVENTORY_Price

Relationships

Customer - Invoice

- One-to-Many (1:M)
- A customer can have zero or more invoices.
- An invoice is generated for exactly one customer.

Invoice - Invoice_Inventory

- One-to-Many (1:M)
- An invoice can include multiple inventoried items.
- Each invoiced item belongs to one invoice.

Inventory - Invoice_Inventory

- One-to-Many (1:M)
- An inventory item can be included in multiple invoices.

NOTE: Each invoice_inventory references one inventory item.

Data Validation

Primary and Foreign Key Validation

Our team has ensured that the integrity of the database design is upheld through primary and foreign key validation. Specifically, the foreign keys **“**INVOICE_ID” and “INVENTORY_ID” in the “Invoice_Inventory” table are correctly structured to reference the primary keys in their respective parent tables, ensuring referential integrity. This guarantees that every entry in the “Invoice_Inventory” table corresponds to valid records in both the “Invoice” and “Inventory” tables, preventing orphaned records and maintaining data consistency.

Furthermore, to preserve the uniqueness of each order-item relationship, a composite key (INVOICE_ID + INVENTORY_ID) has been implemented within the “Invoice_Inventory” table. This acts as a safeguard against duplicate entries, ensuring that no order contains the same item more than once in a redundant manner and enhances the reliability of the database.

Normalization

The database schema has also been normalized to the Third Normal Form (3NF), ensuring a well-structured and efficient data model. In accordance with 3NF, all non-key attributes are fully functionally dependent on the primary key, eliminating any partial dependencies that could lead to redundancy or inconsistencies.

Each entity is clearly defined with appropriately assigned attributes, reducing the risk of anomalies during data modification operations such as inserts, updates, or deletions. This will also ensure the resulting database is capable of efficiently handling complex queries.

CRUD Matrix Table

CRUD Matrix Table

Entity / Process	ALL_CUSTOMERS_VIEW	ALL_ORDERS_VIEW	ALL_ORDERED_ITEMS_VIEW	CUSTOMER.InsertCustomer	CUSTOMER.UpdateCustomerInfo	CUSTOMER.DeleteCustomerByID	ORDER.PlaceOrder	ORDER.CancelOrder
CUSTOMER	R			C	U	D		
ORDER		R					C	D
ORDERED_ITEM			R				C	D
ITEM			R					

CRUD Matrix Table continued...

Entity / Process	ORDER.UpdateOrderDate	ORDERED_ITEM.UpdateQuantity	ITEM.InsertNewItem	ITEM.UpdateItemInfo	ITEM.DeleteItemByID	CUSTOMER_ORDER_SUMMARY	ORDER_DETAILS_BY_ITEM
CUSTOMER						R	
ORDER	U					R	R
ORDERED_ITEM		U				R	R
ITEM			C	U	D		R

Part 4 - User Interface Design

User Interface

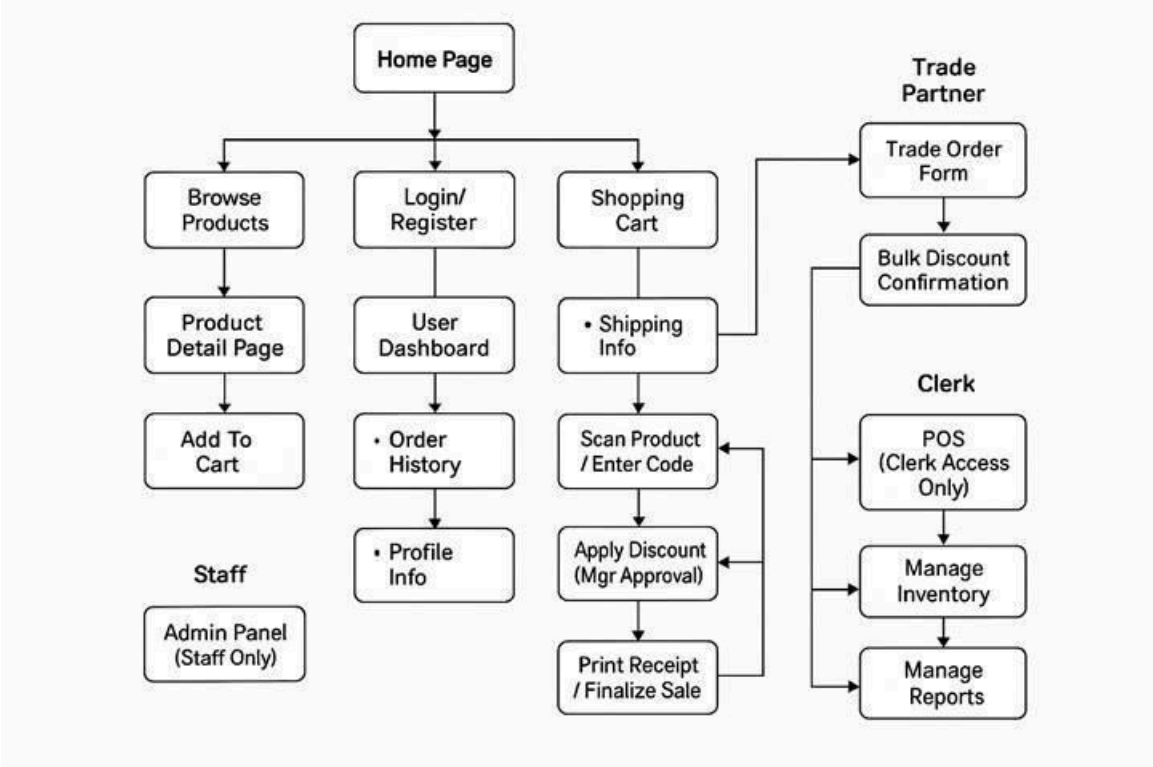
User Profiles

Emma Rodriguez - Retail Customer		
Age: 28	Occupation: Fitness Instructor	Tech Comfort: High
Background:	Shops online frequently, health-conscious, values quality and speed	
Goals:	● Easily browse and order bike accessories ● Track orders in real time	
Pain points:	● Gets frustrated with confusing layouts or hidden shipping info	
Behavior:	● Places small orders frequently ● Accesses the website on her phone	
Expectations:	● Clean, mobile-friendly interface ● Fast checkout with multiple payment options	

Jim Wallace - Trade Partner		
Age: 45	Occupation: Owner of “Jim’s Outdoor Sports”	Tech Comfort: Moderate
Background:	Places monthly bulk orders for resale	
Goals:	● Order large quantities with applied discounts ● Receive accurate invoices and clear payment terms	
Pain points:	● Manual paperwork and delayed invoicing slow down his business	
Behavior:	● Orders via desktop ● Logs in once or twice a month	
Expectations:	● Easy-to-navigate order forms ● Automated invoices and reliable status updates	

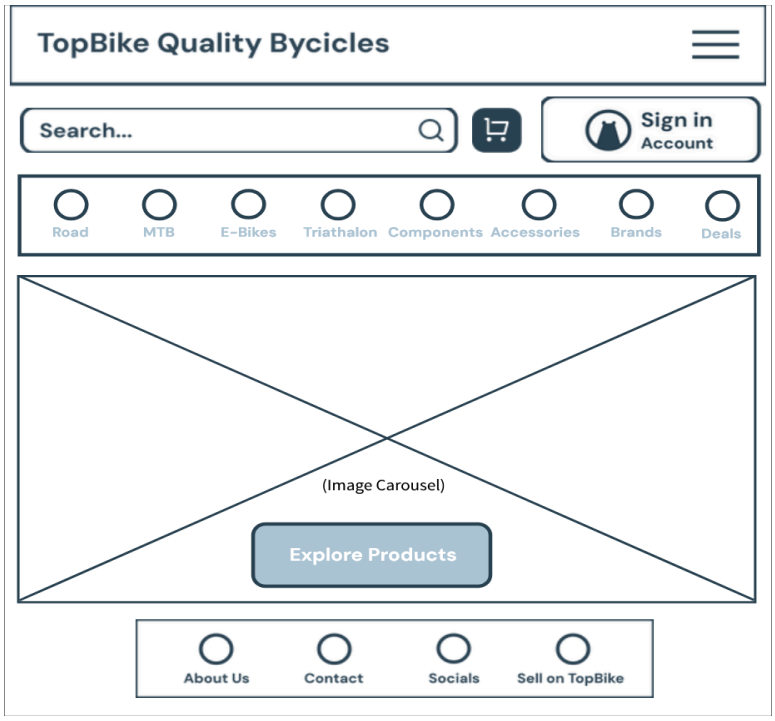
Mia Patel - TopBike Store Clerk		
Age: 22	Occupation: Part-time clerk	Tech Comfort: Moderate
Background:	Handles point-of-sale and customer inquiries	
Goals:	● Process sales quickly at checkout ● Update stock in real-time	
Pain points:	● Hates when the POS crashes or when the interface is cluttered	
Behavior:	● Uses touch screen POS daily	
Expectations:	● Simple layout with clear product info ● Big buttons, fast response time	

Interface Structure Diagram

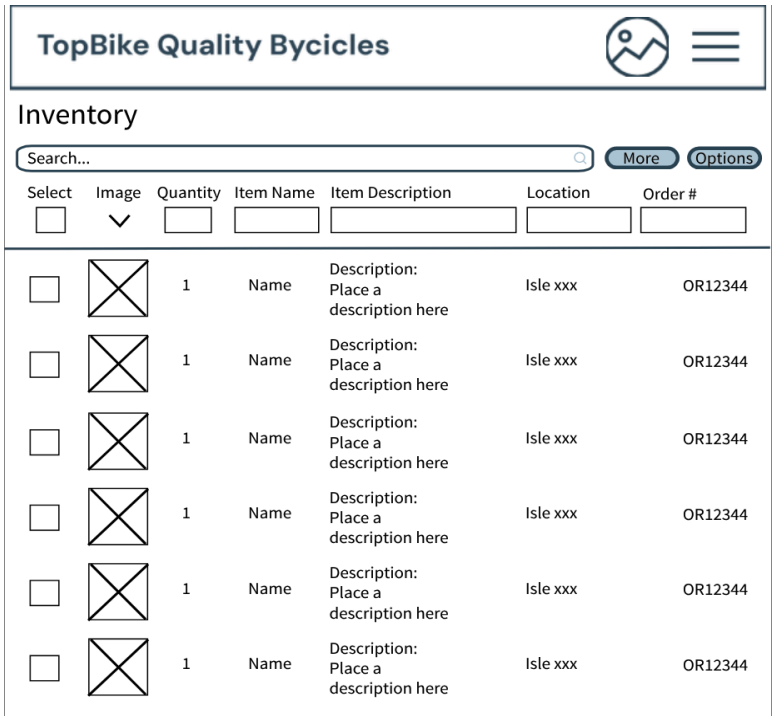


UI Prototypes

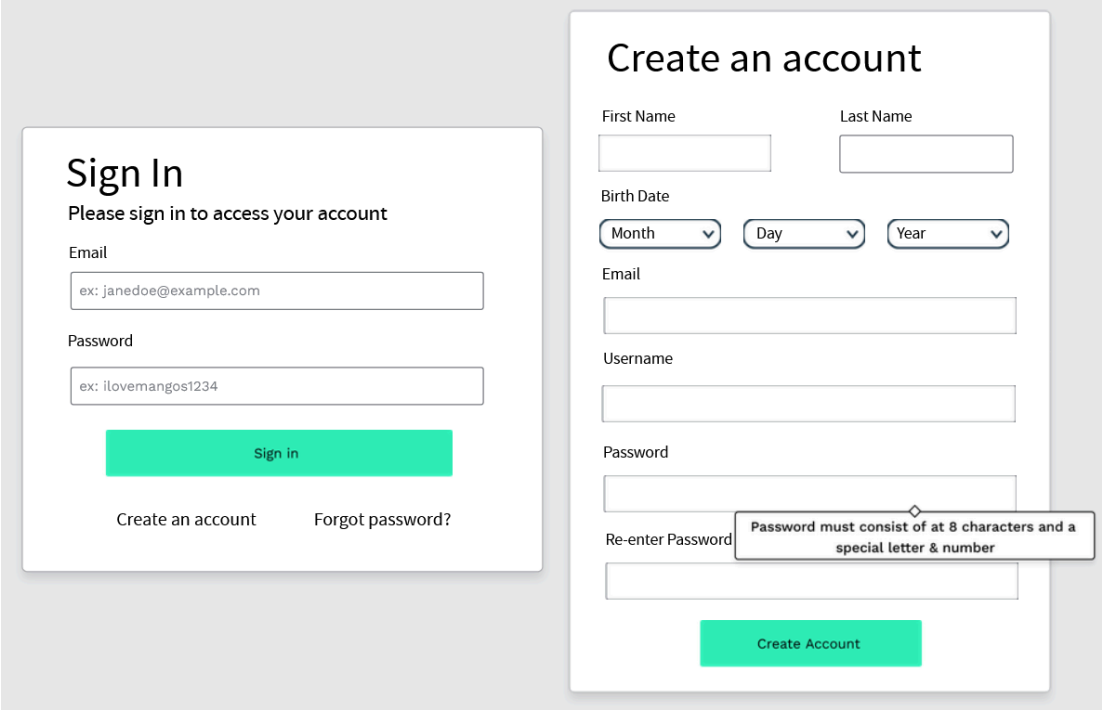
Homepage (Wireframe)



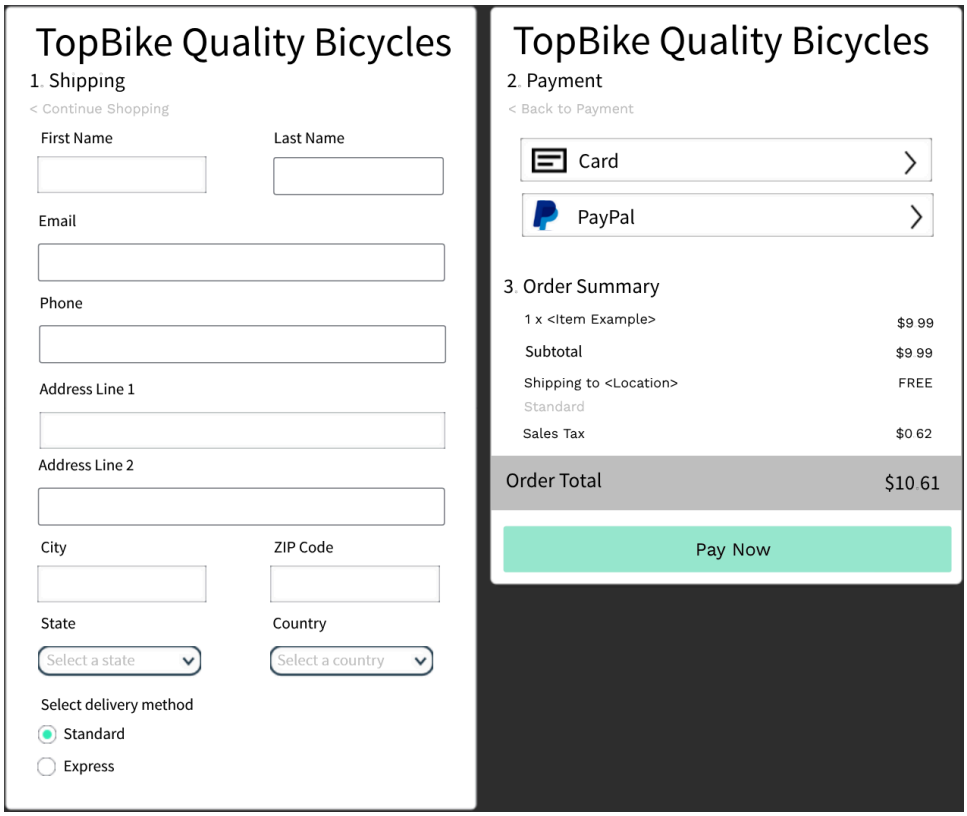
Inventory Management (Wireframe)



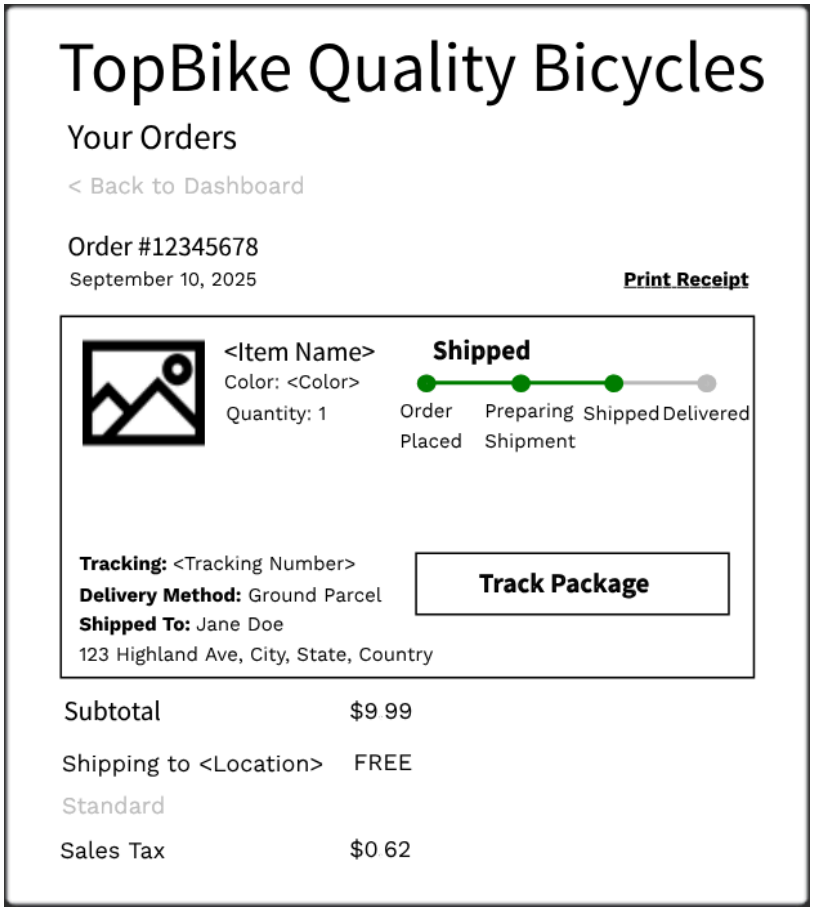
Log-in / Sign Up (Form)



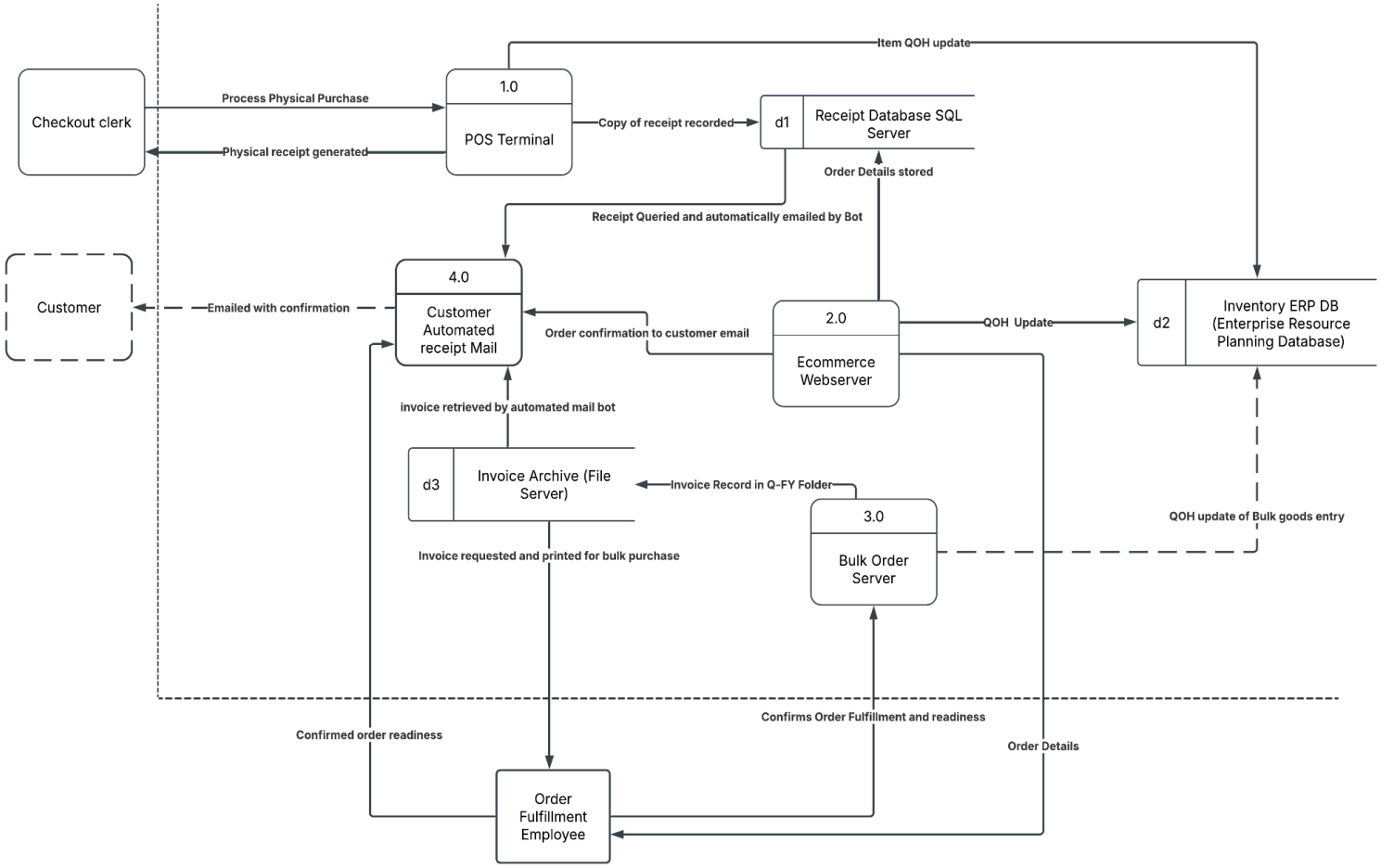
Checkout (Form)



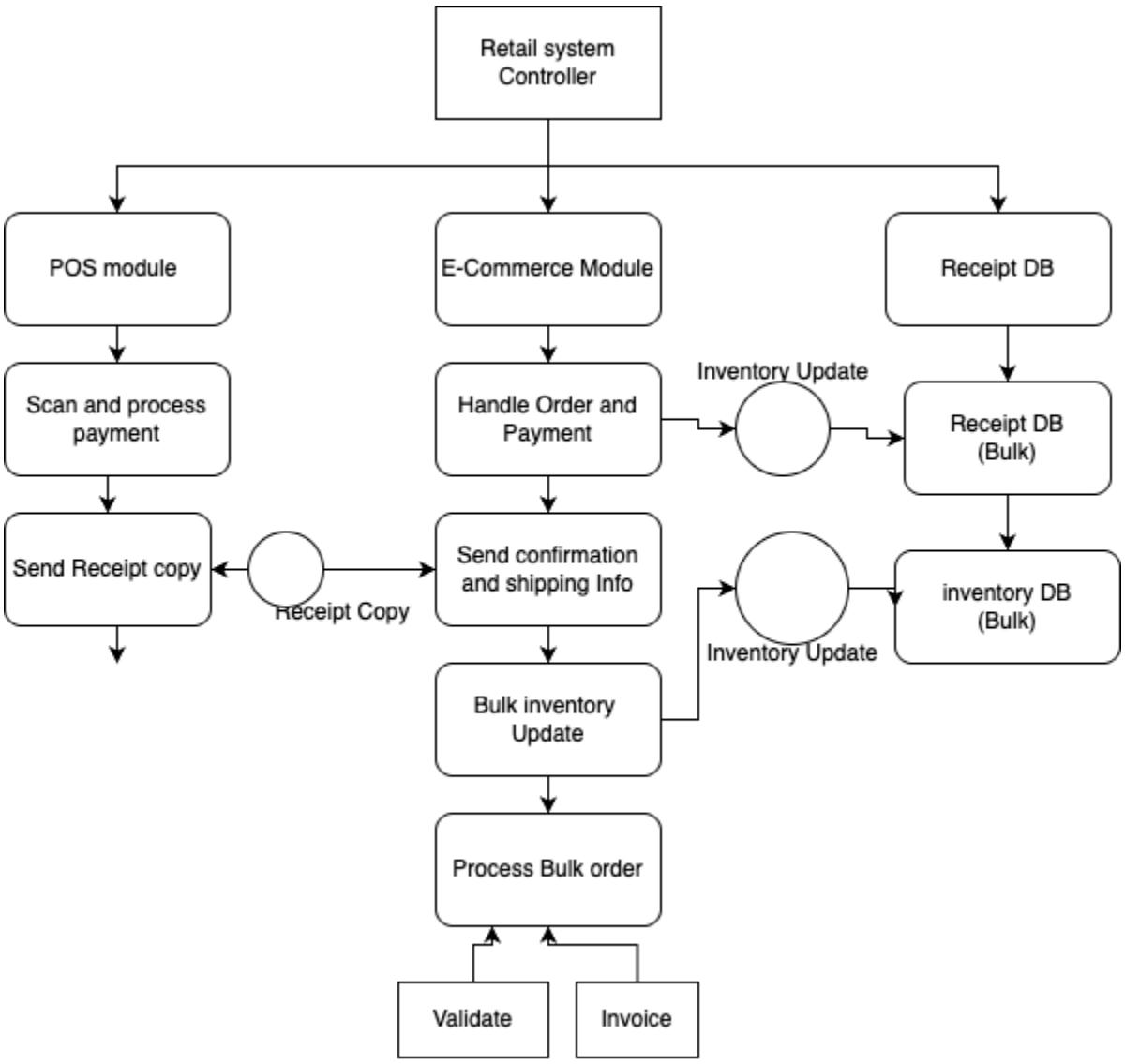
Order details (Report)



Physical Data Flow Diagrams



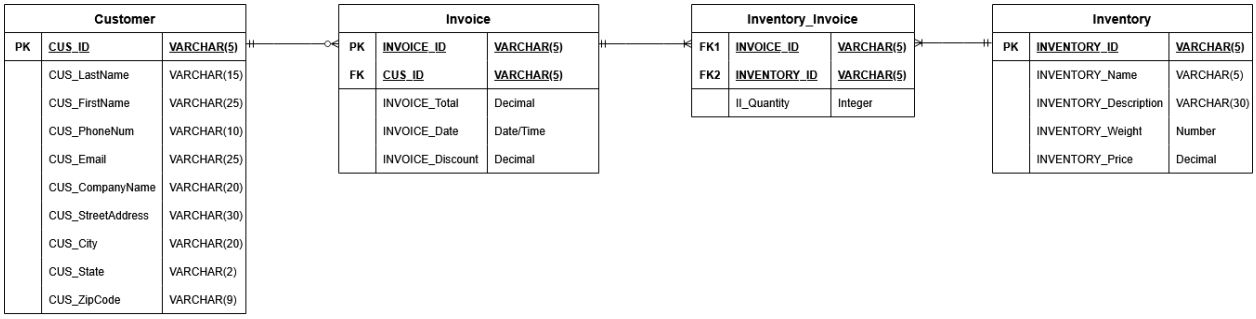
Structure Chart



Part 5 - Test Plan

Physical ERD

Diagram



Overview

Not a lot has radically changed from the logical data model in part 3: the key points that have been changed are the inclusion of primary and foreign key identifiers in the model itself, as well as adding fields with their respective data types. Otherwise, the relationships, cardinality and core entities have remained the same.

Primary and foreign key data types were restricted to a length of 5 for simplicity and storage efficiency. Also of note: some of the data type fields have notably restricted lengths, such as the “CUS_State” and “CUS_ZipCode” fields—this is simply because state can be commonly represented in a 2-letter abbreviation, and ZIP codes never exceed 9 digits in length.

Test Plans

Point-of-Sale (POS) System

Test Plan: Point of Sale

Program ID: TopBike_PointofSale **Version number:** 1.0

Tester: _____ **Date designed:** May 7, 2025 **Date conducted:** _____

Results: ☐ Passed ☐ Open items: _____

Test ID: POS-01

Requirement addressed: REQ-POS-001: Customers must be able to purchase products in-store using the POS terminal.

Objective: Verify end-to-end customer purchases via POS, including item scanning, payment handling, inventory updates, and receipt generation.

Test cases

Interface ID	**Data Field**	**Value Entered**
POS Terminal	Product Barcode	"1234567890123"
POS Terminal	Product ID	"INVALID123"
Payment Window	Payment Method	"Credit Card"
Payment Window	Cash Amount	"$20 on $14.99"
POS Terminal	Cancel Transaction	"Clicked"
System Restart	Recovery Mode	"Restart POS mid-transaction"

Script

1. Launch POS terminal application.
2. Scan a valid product barcode (1234567890123).
3. Manually enter an invalid product ID (INVALID123).
4. Select credit card as payment method and complete transaction.
5. Run another transaction, this time with $20 cash for a $14.99 purchase.
6. Attempt to cancel a transaction before finalizing payment.
7. Simulate a system crash/restart and observe if the session clears and logs the event.

Expected results/notes

- Step 2: Product appears with correct name and price in cart.
- Step 3: Error message displayed; product not added.
- Step 4: Credit card payment completes and receipt is printed.
- Step 5: Correct change returned; receipt includes item and amount paid.
- Step 6: Cart is cleared and transaction aborted successfully.
- Step 7: Crash is logged; upon restart, previous session is not resumed.

Online Ordering System

Test Plan: Online Ordering

Program ID: TopBike_OnlineOrdering **Version number:** 1.0

Tester: _____ **Date designed:** May 7, 2025 **Date conducted:** _____

Results: ☐ Passed ☐ Open items: _____

Test ID: ORD-01

Requirement addressed: REQ-ONL-001: Customers must be able to place orders via the website.

Objective: Verify end-to-end online ordering, from product search through payment processing and order confirmation.

Test cases

Interface ID	Data Field	Value Entered
Home Page	Search Field	“Mountain Bike Pro”
Product Details Page	Quantity Selector	“2”
Cart Page	Coupon Code	“SPRINGSALE2025”
Checkout Page	Shipping Address	“456 Elm St, Anytown, USA”
Payment Page	Credit card number	“4111 1111 1111 1111”
Confirmation Page	Email Address	“John.Smith@example.com”

Script

1. Navigate to https://topbike.example.com.
2. (If required) Log in using valid credentials.
3. On the Home Page, enter “Mountain Bike Pro” in the search bar and submit.
4. From the search results, click the desired product to open the Product Details Page.
5. Set quantity to 2 and click Add to Cart.
6. On the Cart Page, enter coupon code SPRINGSALE2025 and click Apply.
7. Click Proceed to Checkout and enter “456 Elm St, Anytown, USA” as the shipping address.
8. Then, continue to the Payment Page, and enter credit card number 4111 1111 1111 1111, valid expiration date, CVV, and billing address.
9. Submit payment and wait for the Confirmation Page.
10. Verify that the order number appears and that a confirmation email is sent to John.Smith@example.com.

Expected results/notes

- Step 3: “Mountain Bike Pro” appears in search results with correct pricing and availability.
- Step 5: Cart updates to show two units of the selected product, correct line-item price.
- Step 6: Order total reflects the SPRINGSALE2025 discount.
- Step 7: The Shipping address is saved and displayed on the review page.
- Step 8: Payment processes without error; transaction is recorded.
- Step 9: Confirmation Page displays a valid order number, order summary, and estimated delivery date.
- Step 10: Confirmation email arrives within 5 minutes, contains order details and contact info.

Invoice Generation System for Trade Customers

Test Plan: Invoice Generation

Program ID: Topbike_InvoiceGeneration **Version number:** 1.0

Test: _____ **Date designed:** May 8, 2025 **Date conducted:** _____

Results: ☐ Passed ☐ Open items: _____

Test ID: INV-01

Requirement addressed: REQ-INV-01: Invoices should be generated for trade customers only.

Objective: Generate accurate, itemized invoices for trade customers based on purchase orders, including calculations for discounts, taxes, and totals, and export as a PDF. Handle invalid inputs and maintain proper records.

Test cases

Interface ID	Data Field	Value Entered
Invoice Generation Page	Purchase Order ID	"PO12345"
Invoice Generation Page	Trade Customer ID	"T001"
Invoice Details Page	Tax Rate	"7.5%"
Invoice Details Page	Discount Code	"TRADE2025"
Payment Page	Payment Method	"Credit Card"
Payment Page	Card Number	"4111 1111 1111 1111"
Confirmation Page	Export Option	"Generate PDF"
Error Handling Page	Invalid PO ID	"INVALID123"

Script

1. Access the Invoice Generation module.
2. Enter a valid/invalid Purchase Order ID and observe the response.
3. Apply trade customer discount if eligible.
4. Calculate tax and verify totals.
5. Attempt to generate and export an invoice as a PDF.
6. Simulate a system crash during generation and observe the system response.

Expected results/notes

- Step 1: Valid PO = An invoice is generated with accurate itemization, totals, and discounts.
- Step 2: Invalid PO = Error message displayed.
- Step 3: Discount applied correctly for eligible trade customers.
- Step 4: PDF generated with correct formatting and content.
- Step 5: The system recovers from the crash without incomplete records